INCH-POUND

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MILITARY SPECIFICATION

# SWAGING MACHINES, ROTARY, ELECTRIC, FLOOR MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers three types of rotary, electric motor driven, floor mounted, swaging machines with equipment and accessories specified herein.

1.2 Classification. The swaging machine shall be of the following types and sizes listed in tables I, II, and III. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Rotary die swagers - Reduces cross sections to rounds with reciprocating dies revolving around the workpiece, inside a roller cage.

Type II - Stationary die swagers - Forms round or non-round sections on tubing and solid bars, and reduces, tapers, and points ferrous and non-ferrous metal by dies in a stationary spindle with reciprocating action from a roller cage.

Type III - Die closing swagers - Performs cable terminal swaging, center reductions, and parts assembly on round sections by reciprocating wedge controlled dies rotating around the workpiece, inside a roller cage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government,

DRAWINGS

MS-20663A Ball End, Wire Rope, Swaging, Double Shank

MS-20664B Ball End, Wire Rope, Swaging, Single Shank

MS-Z0667C Terminal, Wire Rope, Swaging, Fork End

MS-20668B Terminal, Wire Rope, Swaging, Eye End

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.



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(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) forma part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundanmetal Rating Factors and Calculation Methods for Involute Spur and-Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.7 - Cold Headers and Cold Formers - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA/JIC T2.24.1 - Hydraulic fluid power - Systems standard for industrial machinery.

(Application for copies should be addressed to the National Fluid Power Association, Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted )

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)



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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one conplete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4

3.2 Design The machine shall be new (not a prototype) and one of the manufacturer s current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to met the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall be designed to reduce diameters, or form definite shapes in tubing or solids by swaging.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and netric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate die or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.



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3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electrical, hydraulic, and Pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the swaging machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI 611.7. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.7, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the machine, and its components, the machine shall meet all applicable Environnmtal Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall conply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. An automatic lubricating system shall be provided for all ways, axis screws, parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FEO-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03, AGMA 2001.B88, and AGMA 2000-A88 for the inch-pound system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel,

3.4 Components. The machine shall consist of not less than a base, spindle, flyweheel, drive, roll cage, dies, die closing device, and the other components required to meet requirements specified herein.

3.4.1 Base. The base shall be an iron or steel casting, or a steel fabrication-he base shall support the swaging head, the rotor and absorb and dampen all vibrational loads due to machine operations. Means for securing the machine to the floor shall be provided.

3.4.2 Spindle. A hollow steel spindle shall be mounted in the head with anti-friction bearings of a size, type, and precision class suitable for the application and designed to withstand repeated axial loads. The hammers shall be mounted to the spindle.

3.4.3 Flywheel. The flywheel shall be a casting of iron or steel, dynamically balanced after finishing. The design and mass of the flywheel shall be as required to store and deliver energy required to perform any



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operation within machine capacity without imposing overload on the driving motor. Flywheel slowdown due to energy imparted to perform work shall not exceed 5 percent of no-load speed.

3.4.4 Drive. Drive shall be by electric motor through a driving mechanism to the flywheel mounted directly on the spindle, or to the rotating head.

3.4.5 Roll cage. A roll cage or rack assembly containing bearing steel rollers in recesses shall be mounted in an inside or head ring. On types I and III machines, the cage or rack shall remain free to impart striking force to rotating hammer. On type II machines, the cage or rack shall revolve with the rotating head to impart striking force to stationary hammers,

3.4.6 Dies. Unless otherwise specified (see 6.2), the machines shall be two-die type and shall be furnished with dies as specified (see 6.2).

3.4.7 Die closing device. (Type III machines only. ) A die closing device consisting of a pair of hydraulically actuated wedges interposed between the hammers and dies to provide controlled opening and closing shall be furnished. Hydraulic pressure for the wedge operation shall be generated by a motor driven pump acting through a hydraulic cylinder connected to the wedge rod.

3.4.8 Coolant system. When specified (see 6.2), the coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.9 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of NFPA/JIC T2.24.1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

3.4.10 Electrical requirements The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual Z30/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.10.1 Motors. Motors shall meet the requirements of 3.4.10. Motors shall be rated for continous duty and shall have ball or roller bearings of the sealed and permanently lubrucated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The rotor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.



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3.4.10.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.10.3 Hour inter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of O to 9,999 hours in increments of not greater than 1 hour. The inter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall be not less than those specified in tables I, II, and III for the machine type and size specified.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The machine shall be capable of swaging as specified in 3.6.1, 3.6.2, and 3.6.3 to the requirements listed in tables I, II, and III using 60,000 PSI tensile strength material.

3.6.1 Type I rotary die swagers. Rotary die machines shall be capable of pointing, forming, tapering, and reducing the diameter on tubing, wire, and malleable solids.

3.6.2 Type II stationary die swagers. Stationary die machines shall be capable of pointing and tapering square and rectangular entry cross sections. Type II machines shall form complete fluted shapes, splines, blade ends, and other cross sections from round, square, or rectangular entries.

3.6.3 Type III die closing swagers. Die closing machines shall be capable of assembling USAF - Military Standard (MS) cable terminals and other fittings to cable, and locking banks to pins or shafts. The machine shall form center reductions between shoulders. Type III machines shall be capable of performing as a type I machine in pointing and reducing round entries.

3.6.4 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.7.1 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.



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3.8 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.9 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant,

TABLE I. Type I - Rotary die swagers.

| Sizes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| Capacity, solids | 1/16" | 1/8" | 3/8" | 1/2" | 5/8" | 1" | 1-3/8" | 2" | 3" |
| Capacity tubing | 1/4" | 5/16" | 3/4" | 1" | 1-3/4" | 2" | 3" | 4" | 6" |
| Diameter of hole through spindle (minimum) | 1/4" | 3/8" | 7/8" | 1-5/16" | 1-7/8" | 2" | 3-3/16" | 4-1/4" | 6-1/4" |
| Motor (HP) | 1/4 | 1/3 | 2 | 3 | 5 | 10 | 20 | 30 | 40 |

TABLE II. Type II - Stationary die swagers.

| Sizes | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Capacity, solids | 13/32" | 3/4" | 1-1/2" | 2" | 2-1/2" |
| Capacity, tubing | 1" | 2-1/4" | 3-3/8" | 4-1/2" | 6 |
| Diameter of hole through spindle (minimum) | 15/16" | 2-1/2" | 3-11/16" | 4-1/2" | 6-1/4" |
| Motor HP (mininum) | 3 | 7-1/2 | 15 | 25 | 40 |



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TABLE III. Type III - Die closing swagers.

| Sizes | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Capacity, solids | 3/16" | 3/8" | 1/2" | 1" | 1-1/2" |
| Capacity, tubing | 1/2" | 3/4" | 1" | 2" | 3" |
| Capacity, cable | 1/16" | 1/4" | 5/16" | 3/4" | 1" |
| Standard MS cable terminals: | | | | | |
| Fork end | MS20667-3 | MS20667-8 | MS20667-10 | MS20667-24 | MS20667-32 |
| Eye end | MS20668-3 | MS20668-8 | MS20668-10 | MS20668-24 | MS20668-32 |
| Single shank ball | MS20664-2 | MS20664-8 | MS20664-10 | 1-1/4" dia | NA |
| Double shank ball | MS20663-2 | MS20663-8 | MS20663-10 | 1-1/4" dia | NA |
| Hole through spindle, diameter, inches | 1/2" | 7/8" | 1-1/4" | 2" | 3-3/16" |
| Motor HP | 3/4 | 3 | 7-1/2 | 15 | 20 |

NOTE: Requirements are not less than those shown above, using 60,000 PSI tensile strength material.

3.9.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant and viscosity
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.9.2 Namplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nonmclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture



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3.9.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirments plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2 ], the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.10 Workmanship Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sanpling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.



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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the test(s) in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test.
machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 1040 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. Not less than five tubes and solids of material listed in tables I, or III for the type and size machine specified, equal to maximum capacity of the machine, shall be swaged to the shape of the dies ordered or furnished by the supplier. The finished shapes shall be dimensionally correct. Type III machines shall be further operated to assemble no less than five MS standard fittings to cable. Fittings and cable shall be of the type and size for which dies are procured or are available. The assemblies shall evidence that complete swaging has been accomplished without distortion of the fitting. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.4 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.1. The measurements shall be made at the operators work station and each corner of the machine. Noise emitted by the “machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level inter at slow response. The test shall be conducted with the machine operating under loaded and unloaded conditions.



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4.8.5 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

## 5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Machines procured to this specification are intended for use in pointing and forming tubing, wire, rod, other shaped solids, and assembling fittings to cable and hose.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Noise level, if different (3.2.5.1).

h. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.6).

i. Specify surface finish requirements, if different (see 3.3.3).

j. Painting, if different (see 3.3.5.).

k. Dies, number required, if different (see 3.4.6).



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l. Die requirements, type and size as specified (see 3.4.6).

m. Coolant system, if required, fully describe (see 3.4.8).

n. Voltage requirements, if different (see 3.4.10).

o. Initial voltage, as specified (see 3.4.10).

p. Motors, if different (see 3.4.10.1).

q. Worklight, if required (see 3.4.10.2).

r. Hour meter, if required (see 3.4.10.3).

s. Size and capacity, if different (see 3.5).

t. Specify physical size limitation, if required (see 3.5.1).

u. Additional performance features, if required (see 3.6.4).

v. Specify and fully describe optional equipment, if required (see 3.7.1).

w. Specify applicable repair parts, if required (see 3.8).

x. Lubrication plate, if required (see 3.9.1).

y. Lubrication plate information, if different (see 3.9.1),

z. Namplate, if different (see 3.9.2).

aa. Hydraulic fluid requirements plate, if required (see 3.9.3).

bb. Hydraulic fluid requirements plate information, if different (see 3.9.3).

cc. Inspection conditions, if different (see 4.3).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (4.5).

ff. Acceptance test, if different (see 4.6).

gg. Additional performance tests, if required (see 4.8.3.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of



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the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| N/A | DI-TMSS-80527 | Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English Language |
| N/A | DI-TMSS-80528 | Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals | Require all data be in the English Language |

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior-Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights "and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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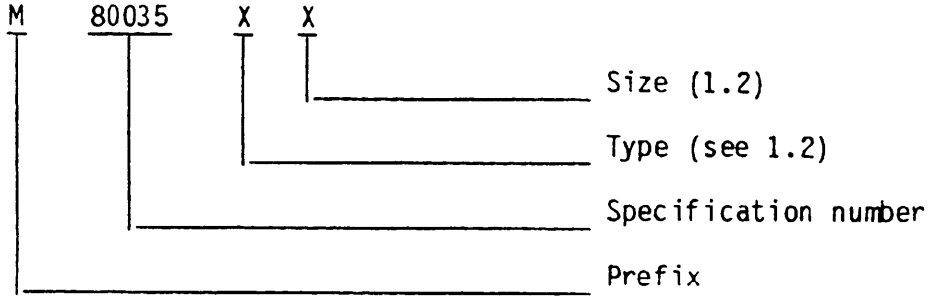
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Nunbers (PINs). The PINs to be used for machines acquired to this specification are created as follows:




6.13 Subject term Key word listing.

Die closing
Roll cage
Rotary die
Swaging

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy -MC, YD
Air Force - 84

User activity:
Navy - OS

Preparing Activity:

DLA - IP

Project (3446-0025)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-S-80035C | 92 02 12 |

3. DOCUMENT TITLE
SWAGING MACHINES, ROTARY, ELECTRIC, FLOOR MOUNTED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial | (2) AUTOVON |
|---|---|---|
| Defense Industrial Plant Equipment Center<br>ATTN: DIPEC-SSG | (901)-775-6694 | 683-6694 |
| c. ADDRESS (Include Zip Code)<br>2163 Airways Blvd.<br>Memphis, TN 38114-5213 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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